MIL-J-16128C(SHIPS)
15 November 1972
SUPERSEDING
MIL-J-16128B(SHIPS)
10 April 1964
(See 6.4)

MILITARY SPECIFICATION

JARS, BATTERY, LAMINATED RUBBER, SUBMARINE

1. SCOPE

1.1 This specification covers laminated rubber jars for use in assembling submarine main storage battery cells.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein.

SPECIFICATIONS

FEDERAL
PPP-B-621 - Boxes, Wood, Nailed and Lock-Corner.

MILITARY
MIL-P-15011 - Pallets, Material Handling, Wood, Past Construction 4-Way Entry.

STANDARDS

FEDERAL
FED-STD-601 - Rubber: Sampling and Testing.

MILITARY
MIL-STD-129 - Marking for Shipment and Storage.
MIL-STD-1186 - Cushioning, Anchoring, Bracing, Blocking and Waterproofing with Appropriate Test Methods.

DRAWINGS

MILITARY
3132750 - Jar, Storage Battery, Laminated Rubber, SARGO.
3247501 - Jar, Storage Battery Cell for GUPPY Mod A or GUPPY Mod C Batteries.
S6204-3112561 - Submarine Battery Cell Jar, Laminated Rubber, Type DRX-47 for SSK Boats.
90262-S6204-941200 - Submarine Storage Battery Cell Jar, Laminated Rubber, GUPPY.

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Material.

3.1.1 Jars shall be made of one of the following materials, as specified (see 6.1):

(a) 100 percent natural rubber with natural soft rubber insert.
(b) 50-50 percent natural-synthetic hard rubber with natural soft rubber insert (overall natural rubber content, 50 percent maximum).

3.1.2 The materials used in the preparation of the hard and soft rubber compounds shall be free from deleterious substances.

3.1.3 The hard rubber shall be free from metallic particles and all granular matter and, with uniform consistency free from lumps or aggregations not thoroughly incorporated into the rubber compound during mixing.

3.2 Hard rubber.

3.2.1 Composition. Iron present as such and in combination shall not exceed 0.3 percent. Manganese present as such and in combination shall not exceed 0.003 percent. The total loss of weight as determined

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by the solubility test (see 4.3.1.2) shall not exceed 4 milligrams per square inch of surface of test piece. The compound shall be such that it will not be subject to embrittlement with age. The incorporation of anti-oxidants in the compound is recommended; however, this shall not be so construed as to permit the use of substances which may have any deleterious effect on lead-acid storage battery operation. Samples of hard rubber shall be subjected to accelerated aging tests (see 4.3.1.5).

3.2.2 Specific gravity. The specific gravity shall be determined as specified in 4.3.1.3.

3.2.3 Tensile strength. The tensile strength shall be not less than 4000 pounds per square inch (p.s.i.) when tested as specified in 4.3.1.4.

3.3 Soft rubber. The soft rubber used for the soft rubber insert shall contain not less than 80 percent raw rubber by volume, and shall have a minimum ultimate elongation of 300 percent (see 4.2.2.4).

3.4 Laminated section.

3.4.1 Transverse deflection. The laminated section shall withstand a load of 140 pounds without breaking and shall show an average deflection of not less than 0.12 inch at the 140 pound load (see 4.3.2.1).

3.4.2 Impact strength. The impact strength shall be not less than 100 foot-pounds (see 4.3.2.2), except that on 10 percent of samples tested a minimum impact strength of 75 foot-pounds shall be permissible.

3.5 Soft rubber foot bars.

3.5.1 Composition. Soft rubber foot bars shall be made from a compound containing no more than 4 percent sulfur by weight (see 4.3.3.1).

3.5.2 Tensile strength. The tensile strength of soft rubber foot-bar material shall be not less than 450 p.s.i. after being subjected to the aging test specified in 4.3.3.3.

3.5.3 Elongation. The elongation of soft rubber foot-bar material shall be not less than 150 percent after being subjected to the aging test specified in 4.3.3.3.

3.6 Jars.

3.6.1 Design and construction. Jars shall be designed and constructed in accordance with Drawings 3132750, 3247501, S6204-3112561 and 90262-S6204-941200.

3.6.2 Dimensions and tolerances. Dimensions and tolerances shall be as shown on Drawings 3132750, 3247501, S6204-3112561 and 90262-S6204-941200.

3.6.3 Jars shall be molded.

3.6.4 Dielectric strength. Jars shall withstand an a.c. potential, measured by the arcing of a 1-1/4 inch spark-gap using needle-point terminals, and applied between the inside and outside surfaces of the jar (see 4.3.4.1).

3.6.5 Hydrostatic pressure. The total permanent set under the hydrostatic test shall not exceed 1 inch. The maximum bulge shall not exceed 2-1/8 inches (see 4.3.4.3).

3.6.6 Stiffness. Jars which show no deflection or jars which bow out less than 3/8 inch at 2000 pounds at 75°F. are considered acceptable. Jars which show no deflection or jars which bow out less than 3/8 inch at 1000 pounds at 115°F. are considered acceptable.

3.6.7 Corner strength. Jar corners shall not crack or rupture when subjected to 150 pounds pressure (see 4.3.4.6).

3.7 Workmanship. The workmanship shall be first class in every respect. All surfaces shall present a neat, smooth, and finished appearance, and shall be free from injurious pitting, fissures, blow-holes, rough spots, rubber scale, blisters, and other deformations which may be caused by the presence of air pockets in the compound, air pockets between laminations, and other air pockets which may be formed between the rubber and the mandrel. Vertical ribs and horizontal bands forming and strengthening the outer walls of jars shall be true with the surface of the jar and shall be solidly joined to each other and to the adjacent surfaces of the jar.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable

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for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Sampling.

4.2.1 Preparation of samples.

4.2.1.1 For testing purposes, each vulcanizer heat of jars (of the same dimensions) shall be represented by one test slab of the same thickness and construction as the jar wall. The test slab shall be prepared from the rubber mixture used in the construction of the jars in the vulcanizer heat and shall be vulcanized with and under the same conditions as the jars of the heat. The test slab shall be of sufficient size to furnish two pieces 6 inches long by 5/8 inch wide, two pieces, 4-3/4 inches long by 7/8 inch wide, one piece 2 by 2 inches, and one piece 10 inches long by 12 inches wide.

4.2.1.2 After test pieces have been prepared from the test slab in accordance with 4.2.2.1 through 4.2.2.5, the remainder of the test slab shall be kept intact for retesting purposes and for use with jars of the vulcanizer heat requiring reheating. In the event that the Defense Contract Administration Services (DCAS) is located at the manufacturer's plant, the test slab remainders shall be kept by the DCAS in a locker provided by the manufacturer in the DCAS office.

4.2.1.3 In each case where it becomes necessary to reheat a jar for the elimination of defects or blemishes, the remainder of the test slab representing the vulcanizer heat containing the jar to be reheated shall undergo heat treatment identical with that to which the jar is subjected. In the event that more than one jar from a heat must be reheated and if the necessary reheating treatments of the jars are not identical, the test slab remainders shall be given treatment identical with that of the jar which is the poorest of the jars to be reheated.

4.2.2 Number of test pieces.

4.2.2.1 Tensile strength and elongation. For tensile strength and elongation tests specified in 4.3.3.2, 5 rectangular test pieces shall be prepared. The dimensions of these pieces shall be 6 inches long by 5/8 inch wide at ends, 1/2 inch wide in center for at least 2 inches of the length, and 1/8 inch thick.

4.2.2.2 Transverse deflection. For the transverse deflection test specified in 4.3.2.1, 5 rectangular test pieces shall be prepared. The dimensions of these pieces shall be 4-3/4 inches long by 7/8 inch wide by the thickness of the jar wall section, including the soft rubber insert.

4.2.2.3 Solubility test. Two pieces 2 inches long by 2 inches wide by 7/32 inch thick shall be prepared from the test slab for the solubility test specified in 4.3.1.2. This test need not be conducted on every vulcanizer heat of jars but shall be conducted at least once a month during production.

4.2.2.4 Chemical analysis of hard rubber. Three pieces 2 by 2 inches by 1/8 inch shall be prepared for the chemical analysis of hard rubber specified in 4.3.1.1 and two pieces 2 by 2 inches by 1/16 inch shall be prepared for the analysis of soft rubber. These tests need only be conducted once on each contract in lieu of on every vulcanizer heat of jars (see 3.3).

4.2.2.5 Impact. Two pieces 10 inches long by 12 inches wide by the thickness of the jar wall shall be prepared for the impact test specified in 4.3.2.2.

4.3 Test.

4.3.1 Hard rubber.

4.3.1.1 Chemical analysis. The total iron and manganese content shall be determined. The presence of other substances which may have any effect on operation of lead-acid storage battery shall also be determined.

4.3.1.2 Solubility. The solubility test shall be conducted as follows: Dry the test pieces for 1 hour at 212° to 212°F. weigh and record weight. The test pieces shall be placed in sulfuric acid of 1.400 specific gravity in such a manner that the acid shall have free action upon the entire surface. The test pieces shall remain in the acid for 100 hours during which time the acid shall be maintained at 120°F. The test pieces shall be removed from the acid and washed in distilled water until the washings are neutral to methyl orange indicator. The test pieces shall then be dried for 1 hour at 212° to 221°F. and weighed again. The loss in weight shall be taken as the solubility in 1.400 specific gravity sulfuric acid. The average of the 2 test pieces shall be taken as a result of the test.

4.3.1.3 Specific gravity. Specific gravity of hard rubber samples shall be determined by the use of a pycnometer, using 95 percent ethyl alcohol to eliminate errors due to air bubbles.

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4.3.1.4 Tensile strength.

4.3.1.4.1 All tests shall be made on the test pieces after immersion in water at a temperature of 65° to 75°F. for at least 1 hour. The test for tensile strength shall be made immediately following this period, in a testing machine approved by the DCAS. The temperature of the test piece shall be maintained between 65° and 75° F. during the test.

4.3.1.4.2 The force required to effect a rupture shall be applied gradually and uniformly. The jaws of the testing machine shall not separate at a rate to exceed 0.6 inch per minute for hard rubber testing.

4.3.1.4.3 The average of 5 test pieces shall be taken as the result of the test. A record shall be kept of the percentage elongation of the test pieces.

4.3.1.4.4 At the discretion of the DCAS, and with concurrence of the manufacturer, only three test pieces need be prepared and tested instead of the 5 test pieces specified in 4.2.2.1 and 4.3.1.4.3. In such cases, the average of three test pieces shall be taken as the result of the tensile strength and elongation tests.

4.3.1.5 Accelerated aging. Aging tests shall be made on samples of hard rubber in accordance with method 7001 of FED-STD-601, under either of the following conditions:

(a) Geer oven - 7 days at 158°F.
(b) Oxygen bomb - 48 hours at 158°F.

The nature and effect of accelerated aging shall be recorded.

4.3.2 Laminated section.

4.3.2.1 Transverse deflection.

4.3.2.1.1 Tests shall be made on the test pieces after immersion in water at a temperature of 65° to 75° F. for at least 1 hour. Immediately following immersion, the piece to be tested shall be placed upon the supporting abutments of the testing machine and shall be maintained at a temperature of 65° to 75°F. during the test. The distance between supporting abutments shall be such that the axes are 3 inches apart and so centered that the head will run centrally between them. The load in all cases shall be applied at a rate not to exceed 0.6 inch per minute.

4.3.2.1.2 The average of five test pieces shall be taken as the result of the test.

4.3.2.1.3 At the discretion of the DCAS, and with concurrence of the manufacturer, only three test pieces need be prepared and tested instead of the five test pieces specified in 4.2.2.2 and 4.3.2.1.2. In such cases the average of three test pieces shall be taken as the result of the transverse deflection test.

4.3.2.2 Impact test.

4.3.2.2.1 Impact tests shall be conducted on an impact testing machine similar to the machine shown on figures 1 to 4, inclusive. The machine shall have a plunger 3 inches long, made in the shape of a rod cut in half lengthwise. The part of the plunger in contact with the test piece shall have a radius of 1 inch and the weight of the plunger shall be between 1-3/4 and 2 pounds. The plunger shall strike the test piece horizontally across the grain of the rubber.

4.3.2.2.2 The 10 by 12-inch test piece shall be supported in a horizontal plane by steel supports spaces 6-1/2 inches apart so constructed that the test piece can be placed in only one position.

4.3.2.2.3 A 50-pound weight shall be dropped on the plunger from heights corresponding to 30, 40, 75, 100 and 150 foot-pounds. The mechanism attached to the tester shall trap the weight on the rebound thereby permitting it to strike the plunger only once.

4.3.2.2.4 Tests shall be conducted at room temperature ranging from 75° to 85°F. (80°F. average).

4.3.2.2.5 Tests shall be conducted on samples, having no artificial aging, approximately 24 hours after cure.

4.3.2.2.6 An average of two samples shall be taken as the result of this test if the first test sample is less that 100 pounds.

4.3.3 Soft rubber.

4.3.3.1 Chemical analysis. The sulfur in the soft rubber foot bars shall be determined for compliance with 3.5.1.

4.3.3.2 Tensile strength and elongation. Tensile strength and elongation shall be determined in accordance with methods 6111 and 6121 of FED-STD-601 except as specified (see 4.3.1.4.1).

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4.3.3.3 Accelerated aging. Aging tests shall be conducted as specified in 4.3.1.5 once every 90 days or once for every 800 jars manufactured under the same contract.

4.3.4 Jars.

4.3.4.1 Dielectric test. Each jar shall be subjected to the following dielectric test: Place the jar to be tested in a suitable container, fill jar and container with water so that water level is within 1-1/2 inches on the top of the jar, inside and outside; allow jar to soak for at least 5 hours. At the end of this period place one lead from a high-tension transformer into water in the container and place the other lead from transformer into water in the jar. Subject the jar to the test potential continuously for 1 minute. The transformer capacity shall be not less that 1 kw. The air-gap needle-point terminals shall be 1-1/4 inches apart. A needle-point calibration of the testing equipment shall be made before and after each lot of jars is tested to insure that the equipment produces a potential equivalent to a 1-1/4-inch spark gap at the jar. Jars which break down shall not be offered for delivery.

4.3.4.2 Inside dimensions. After checking the outside dimensions of the jar, the inside dimensions shall be checked. For this purpose, the supplier shall furnish a suitable plug gage, the dimensions of which shall be 1/32 inch less than the minimum inside dimensions of the jar. This plug gage shall be checked by the DCAS to see that it is of the proper size.

4.3.4.3 Hydrostatic tests (see 4.4).

4.3.4.3.1 This test shall be performed with the jar in an unwedged condition, that is, not supported along its sides. Caliper jar and record measurements; suspend a weight in the jar in the same manner that the element is suspended in the finished cell. This test weight shall be 25 percent in excess of the weight of the element. In addition to the test weight, fill the jar with water up to the operating level and keep it standing in this condition for 6 hours. During the test the water in the jar shall be maintained at a temperature of 120°F. and the room temperature shall not be allowed to go below 70°F.

4.3.4.3.2 At the end of this period with weight and water in the jar at 120°F. caliper measurements shall be taken on the outside center of the jar on both length and width to determine the point and amount of maximum bulge. Weight and water shall be removed and, after the jar has assumed the temperature of the room, caliper measurements shall again be taken on the outside center of the width and length of the jar to determine the point and amount of permanent set.

4.3.4.3.3 While removing weight and water from jar and while jar is still warm, care shall be taken that the jar is not laid on its side, as this treatment will tend to press the bulge back and affect the final permanent set.

4.3.4.3.4 Jars subjected to the hydrostatic test shall be limited to five jars as a test of design or type, and shall be scrapped at the end of the test.

4.3.4.4 Surface examination. Following the dielectric test, (see 4.3.4.1) a minute examination of the jars shall be made and depressions, holes, pits and other imperfections noted. Blemishes due to the physical characteristics of hard rubber will be permitted, provided the serviceability of the product is not affected. Imperfections due to careless workmanship, the use of neglected mandrels, molds or faulty material will not be permitted. In measuring depressions, holes, and pits, a depth gage shall be used, and the surface surrounding any depression, hole or pit may be removed to facilitate examination.

4.3.4.4.1 Pits or chips.

4.3.4.4.1.1 No pits or chips, either on the inside or outside of jar wall proper, shall be filled or patched, and evidences of any attempt to patch the body of the jar shall cause rejection.

4.3.4.4.1.2 Any pit or chip on the interior or exterior surface deeper than 3/32 inch shall cause rejection. Any pit on the interior surface shall be at least 1 inch distant from a pit on the exterior face of the same wall of the jar. More than five pits or chips on any one interior or exterior face deeper than 1/32 inch shall cause rejection. Any pit or depression on the interior end wall below the element-support ledge wider than 1/4 inch, measured parallel with the element-support ledge, shall cause rejection. This requirement is to prevent moss from collecting in this cavity and short-circuiting adjoining plates. With this possibility in view, all holes, pits, or depressions on the end wall below the element-support ledge shall be ground smooth.

4.3.4.4.2 Blisters.

4.3.4.4.2.1 Jars are made up of three distinct laminations, that is, a layer of soft rubber between two layers of hard rubber. The hard rubber laminations are each made up of several thin layers of rubber plied to the desired thickness on a calender. Blisters in the jar wall can be classified into two general categories. In one category, the blister is formed between two of the three main laminations which make up the jar wall. Generally these blisters are large and not easily broken. In the other category, the blister is formed between the thin, plied layers of a hard rubber lamination, resulting in smaller blisters which, in most cases, are easily broken.

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4.3.4.4.2.2 Each jar shall be thoroughly examined for blisters. Blisters shall be tapped lightly to insure that the blisters have not caused a crack in the lamination. It is important that blisters shall not break easily. IN NO CASE SHALL BLISTERS BE GROUND DOWN OR PRESSED BACK INTO THE WALL. The total blistered area on any one combined inner and outer wall or combined inner and outer bottom shall nor exceed 15 square inches. The total blistered area per jar shall not exceed 30 square inches. The area of any one blister shall not exceed 7-1/2 square inches (approximately 3 inches in diameter), and the projection or bulge of any one blister shall not exceed 3/32 inch.

4.3.4.4.2.3 Jars having the following imperfections shall not be offered for delivery:

(a) Jars with blisters adjacent to the wedging ribs if the blisters interfere with the seating of the wedge.
(b) Jars with blisters on the inner wall which interfere with the raising or lowering of the element test plug.
(c) Jars with blisters in the corners or heavy sections where their extent is difficult to determine and where the structure strength of the jar is affected.
(d) Jars containing one or more broken blisters, leaving a pit deeper than 3/32 inch.
(e) Jars with blisters exceeding the areas specified in 4.3.4.4.2.2.
(f) Jars containing a hard blister exceeding 1/32 inch in height on the inner wall which coincides with a blister exceeding 1/32 inch in height on the outer wall on the same side or end of the jar.
(g) Jars with any soft rubber lamination exposed.

4.3.4.4.3 Separations.

4.3.4.4.3.1 In top of jar. If the separations are more than 1 inch deep, the jar shall be rejected. If the separations are less than 1 inch deep, the separations shall be filled in with compound. A dielectric test shall be carefully made around the separation to reveal latent and invisible cracks.

4.3.4.4.3.2 Between element-support ledge and main body of the air. Separations between the element-support ledge and the main body of the jar shall not exceed 1/16 inch in width. The length of the separation shall not exceed 2 inches for each side of the jar and the depth shall not exceed 1/8 inch. Separations shall not be filled in.

4.3.4.4.3.3 Between adjoining walls of jars in corners. If the separations are more than 1/16 inch deep, the jar shall be rejected. The corners and inside edges of the jar shall be very carefully examined, and the jar shall be given a careful dielectric test at these places. Any separations shall be not more than 1-1/2 inches in length. Such a defect is probably due to rubber not flowing properly, but if it is due to careless workmanship, it shall be discouraged by rejection.

4.3.4.4.3.4 Between the frame bars at the bottom of the jar, and between the side walls and the bottom of the jar. Jars with cracked frame bars shall be rejected. If the separation between the frame bars at the bottom of the jar, and between the side walls and the bottom of the jar extends deeper than 3/16 inch, the jar shall be rejected. Separations shall not be filled in.

4.3.4.4.4 Projections torn from soft rubber foot bars. Replacement with a complete new soft rubber foot bar will be permitted, provided that rubber is live after curing.

4.3.4.5 Stiffness test (see 4.4).

4.3.4.5.1 The relative stiffness of jars shall be determined by using a spring load test fixture (see figure 5). This tester shall be applied to laminated rubber jars to determine the pressure required to buckle the jars between the ribs, particularly at the top. The spring used shall be of the high pressure medium deflection type, 4 inches long with a 2-1/2-inch hole for a 1-inch diameter rod. The spring shall provide 2100 pounds maximum pressure with 260 pounds required for 1/8 inch deflection (maximum deflection 1 inch). Shims and slotted wooden wedges (2 to 3 inches long) shall also be used for this test.

4.3.4.5.2 Two jars shall be assembled with close fitting covers and placed together with the long side adjacent. A spring test fixture (see 4.3.4.5.1) shall be placed between the two jars at the top and adjusted to zero load. Shims shall be inserted between jar and fixture where necessary to make up any difference in jar lengths. Special wooden wedges shall be lightly pressed from the top between each adjacent pair of vertical ribs on opposite jars.

4.3.4.5.3 Cold test. Jars shall be tested with the spring type tester at a room temperature of approximately 75°F. The actual room temperature shall be read and recorded. The load shall be slowly applied on the spring tester and the effect of pressure on the top of the jars shall be determined by the compression of the springs which shall be measured and recorded to the nearest 1/64 inch. Additional load shall be applied in increments of 255 to 260 pounds, with a pause of 2 minutes during which the length of the spring shall be measured. This shall be continued until failure takes place or until the maximum compression of the spring is reached.

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4.3.4.5.4 Hot test. Two jars shall be assembled on a truck. The room temperature shall be read and recorded. The load shall be slowly applied until the spring is compressed 1/2 inch which is equivalent to appoximately 1040 pounds. Without disturbing the jars on the truck, they shall be pushed into an oven which has previously been heated to 115°F. Jars shall be examined occasionally and the fixture shall be tightened to replace any load lost due to deformation of jars. Temperature readings shall be taken and additional heat shall be applied as necessary to keep the temperature within the range of 110° to 115°F. for approximately 2-1/2 hours. Jars not failing under these conditions of tests shall have additional load gradually applied while in the oven until failure takes place or the end of the adjustment on the fixture is reached.

4.3.4.6 Corner strength test. (see 4.4).

4.3.4.6.1 Corner strength test. This test shall be conducted at room temperature, and with the jar lying on one long side but otherwise unsupported. Pressure of 150 pounds shall be applied 3-1/4 inches from each jar corner on the long side by means of a hydraulic jack (see figure 6). A hard rubber section the same thickness as the jar wall shall be used as partial support for the jack. After tests have been conducted on two corners with the jar lying on one side, the jar shall be turned over and tests repeated so that all four corners are tested.

4.3.4.6.2 Jars subjected to the corner strength test shall be limited to five jars as a test of design or type and should be scrapped at the end of the test.

4.4 Tests specified in 4.3.4.3, 4.3.4.5 and 4.3.4.6 shall be performed only at the time a new design of jar is purchased for the first time or when a supplier is making an existing jar type for the first time.

4.5 Retests.

4.5.1 When jars and the representative test-piece slab (see 4.2.1.2 and 4.2.1.3) have been reheated for any purpose, test pieces shall be prepared from the slab and subjected to tensile strength elongation transverse deflection, and impact tests (see 4.3.1.4, 4.3.2.1, and 4.3.2.2).

4.5.2 When jars are reheated for any prupose, they shall again be subjected to the dielectric test (see 4.3.4.1).

4.6 Each jar of a vulcanizer heat represented by a sample which fails to pass the specified chemical analysis, or the tests for tensile strength, elongation, impact strength and transverse deflection shall be rejected. All the jars in a vulcanizer heat which are subjected to reheating to remove defects or blemishes shall be represented by the reheated test-slab remainder (see 4.2.1.3).

4.7 Inspection for preparation for delivery. The packaging, packing and marking shall be inspected for compliance with section 5 of this document.

5. PREPARATION FOR DELIVERY

(The preparation for delivery requirements specified herein apply only for direct Government procurements. For the extent of applicability of the preparation for delivery requirements of referenced documents listed in section 2, see 6.3)

5.1 Packing.

5.1.1 Level A. Unless otherwise specified (see 6.1), battery jars shall be palletized for shipment in accordance with figure 7, type SARGO II, type GUPPY I, type GUPPY I MOD A, or GUPPY I MOD C, as specified (see 6.1).

5.1.1.1 When specified (see 6.1), battery jars shall be packed in overseas type nailed wood boxes conforming to PPP-B-621. Jars shall be padded and cushioned within containers in accordance with MIL-STD-1186.

5.1.2 Level B. Unless otherwise specified (see 6.1), battery jars shall be palletized for shipment in accordance with figure 7, type SARGO II, type GUPPY I, type GUPPY I MOD A, or GUPPY I MOD C as specified (see 6.1).

5.1.2.1 When specified (see 6.1), battery jars shall be packed in domestic type nailed wood boxes conforming to PPP-B-621. Jars shall be padded and cushioned within containers in accordance with MIL-STD-1186.

5.1.3 Level C. Battery jars shall be palletized for shipment on standard commercial pallets in accordance with commercial practice. Palletized jars shall be acceptable by common carrier and shall be satisfactory for safe delivery and acceptance at destination.

5.2 Marking. In addition to any special marking required by the contract or order, marking for shipment

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shall be in accordance with MIL-STD-129.

6. NOTES

6.1 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Material required (see 3.1.1).
(c) Whether palletizing shall be for SARGO type or GUPPY type (see 5.1.1 and 5.1.2).
(d) Packing required (see 5.1.1.1 and 5.1.2.1).

6.2 Type of ships for which the drawings listed in section 2 are applicable:

3132750 - Fleet type (Sargo), Guppy 1A and Guppy 2A.
S6204-3112561 - SSK 1 Class.
90262-S6204-941200 -} Guppy II, SS 563 Class
3247501 -} and SSR 312 and 419.

6.3 Sub-contracted material and parts. The preparation for delivery requirements of referenced documents listed in section 2 do not apply when material and parts are procured by the supplier for incorporation into the equipment and lose their separate identity when the equipment is shipped.

6.4 THE MARGINS OF THIS SPECIFICATION ARE MARKED "#" TO INDICATE WHERE CHANGES (ADDITIONS, MODIFICATIONS, CORRECTIONS, DELETIONS) FROM THE PREVIOUS ISSUE WERE MADE. THIS WAS DONE AS A CONVENIENCE ONLY AND THE GOVERNMENT ASSUMES NO LIABILITY WHATSOEVER FOR ANY INACCURACIES IN THESE NOTATIONS. BIDDERS AND CONTRACTORS ARE CAUTIONED TO EVALUATE THE REQUIREMENTS OF THIS DOCUMENT BASED ON THE ENTIRE CONTENT IRRESPECTIVE OF THE MARGINAL NOTATIONS AND RELATIONSHIP TO THE LAST PREVIOUS ISSUE.

Preparing activity:
Navy - SH
(Project 6140-N352)

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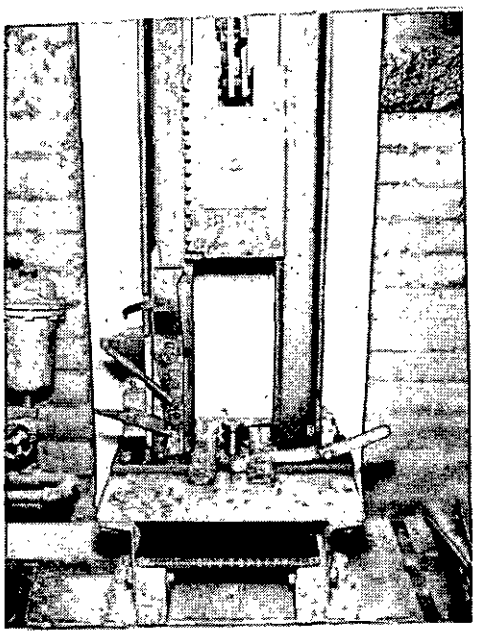


Figure 1 - Impact tester for jar slabs showing plunger raised and test piece in position with weight in position to be dropped.

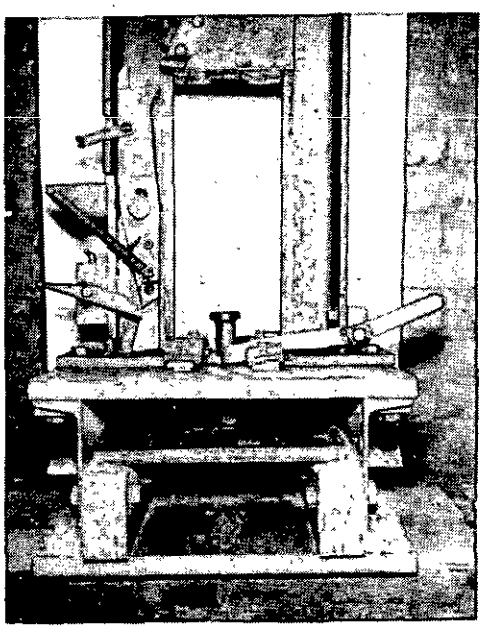


Figure 2 - Impact tester for jar slabs showing test piece in position with plunger ready for impact blow.

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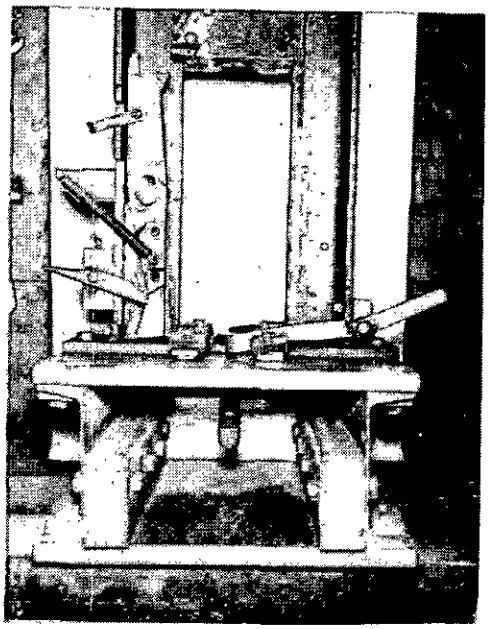


Figure 3 - Impact tester for jar slabs showing plunger down and test piece removed.

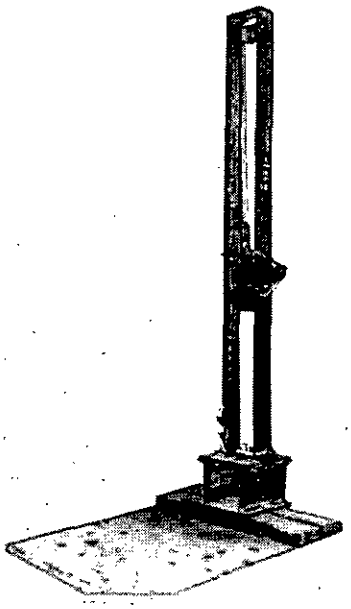
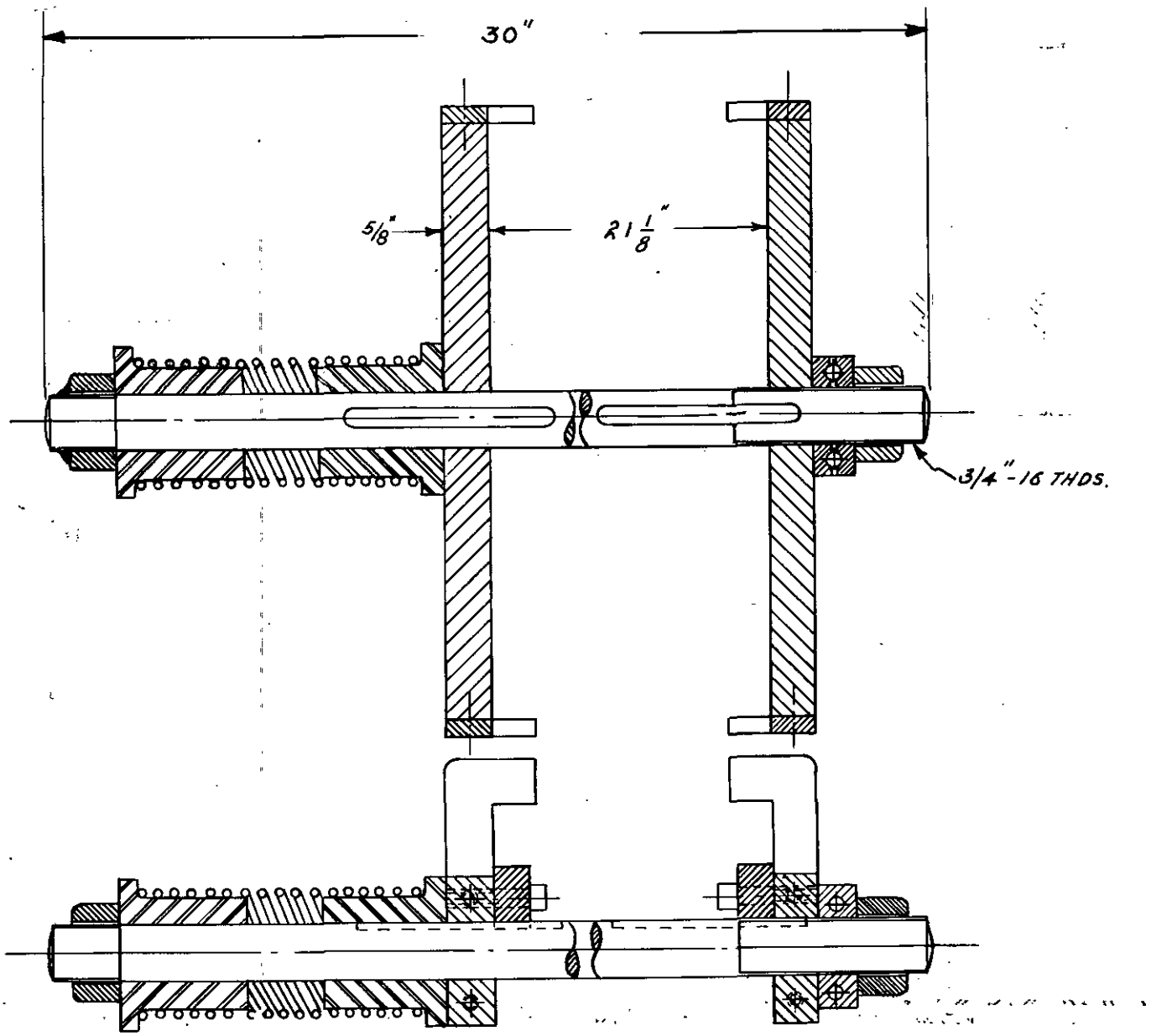


Figure 4 - Impact tester for jar slabs.

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All decimal dimensions to have a limit of ± 0.0625 inch unless otherwise stated.
If fractional figures are given they should be within ± 1/16 inch.

Figure 5 - Spring load test fixture

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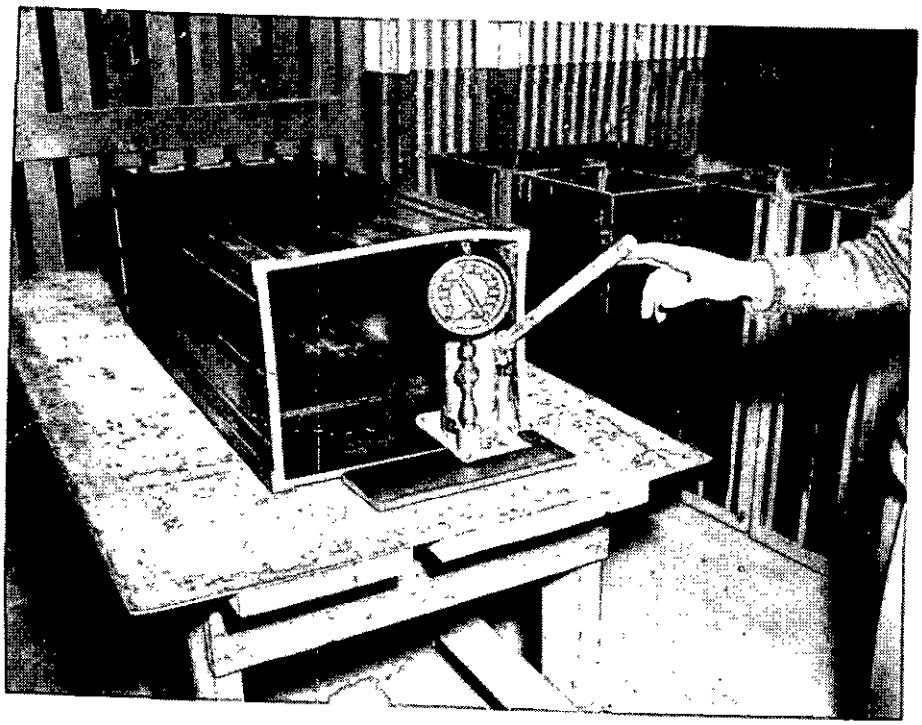
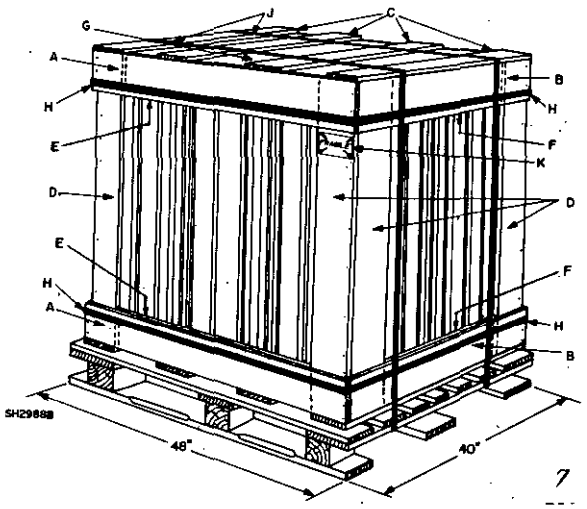


Figure 6 - Hydraulic jack.

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NOTES:

1. Six jars to a pallet.
2. Pallets shall be constructed in accordance with MIL-P-15011, Class 1.
3. Lumber for pallets shall conform to the requirements of MIL-P-15011. All other lumber shall be first-class commercial crating grade and shall be capable of withstanding shipping, storage (including superimposed loading) and reshipment without the necessity of repacking.
4. Vertical frame pieces (D) must fit inside or top and bottom caps. The length of these pieces shall be the height of the tallest jar plus 3/8 inch (allowing 1/8 inch for fiberboard and 1/4 inch for clearance).
5. Filler pieces (E) and (F) installed in top and bottom. Nail filler pieces (E) and (F) in place to top end and top side slats (A) and (B).
6. Two fragile labels shall be affixed to top of diagonally opposite vertical frame pieces.

Bill of materials

Item	Description	Quantity	Size		
			Type SARGO II 1/	Type GUPPY I	Type GUPPY I MOD A or C 2/
A	Side Slats	4	1 by 6 by 49-15/16	1 by 6 by 46-3/4	1 by 6 by 46-1/2
B	End Slats	4	1 by 6 by 45-1/4	1 by 6 by 37-3/4	1 by 6 by 30-1/2
C	Top and bottom Slats	8	1 by 6 by 47-1/4	1 by 6 by 39-3/4	1 by 6 by 32-1/2
D	Vertical Frame Pieces	8	1 by 6 by 50-5/8	1 by 6 by 41-3/4	1 by 6 by 50-3/4
E	Side fillers	4	1 by 6 by 36-1/4	1 by 6 by 33-1/4	1 by 6 by 33
F	End Fillers	4	1 by 6 by 32	1 by 6 by 24-1/4	1 by 6 by 17
G	Fiberboard	1	47-3/4 by 43	42-3/4 by 35-3/4	28-1/2 by 42-1/2
H	Steel Girth Straps	2	3/4W by 190	3/4W by 176	3/4W by 156
J	Steel Binding Straps	2	3/4W by 212	3/4W by 190	3/4W by 190
K	Fragile Labels	2			

1/ If standard pallets are not available, the lengths of the transverse stringers may be increased from 40 to 47 inches. Spacing between the six deck boards will be increased accordingly, and strapping slots shifted as necessary. If standard pallets are used, 3-1/2 inch side and 1-inch end overhand is permissible.

2/ It will be permissible to use a modified pallet having 35-inch transverse stringers and five deck boards for packing if standard pallets are not available.

Figure 7 - Method of palletizing submarine battery jars.

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NAVAL SHIP ENGINEERING CENTER
CENTER BUILDING - SEC 6124B
PRINCE GEORGE'S CENTER
HYATTSVILLE, MARYLAND 20782

OFFICIAL BUSINESS
PENALTY FOR PRIVATE USE $300

POSTAGE AND FEES PAID
DEPARTMENT OF NAVY

DOD 316

COMMANDER
NAVAL SHIP ENGINEERING CENTER
CENTER BUILDING - SEC 6124B
PRINCE GEORGE'S CENTER
HYATTSVILLE, MARYLAND 20782

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No. 22-R255

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DOCUMENT IDENTIFIER AND TITLE

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*

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